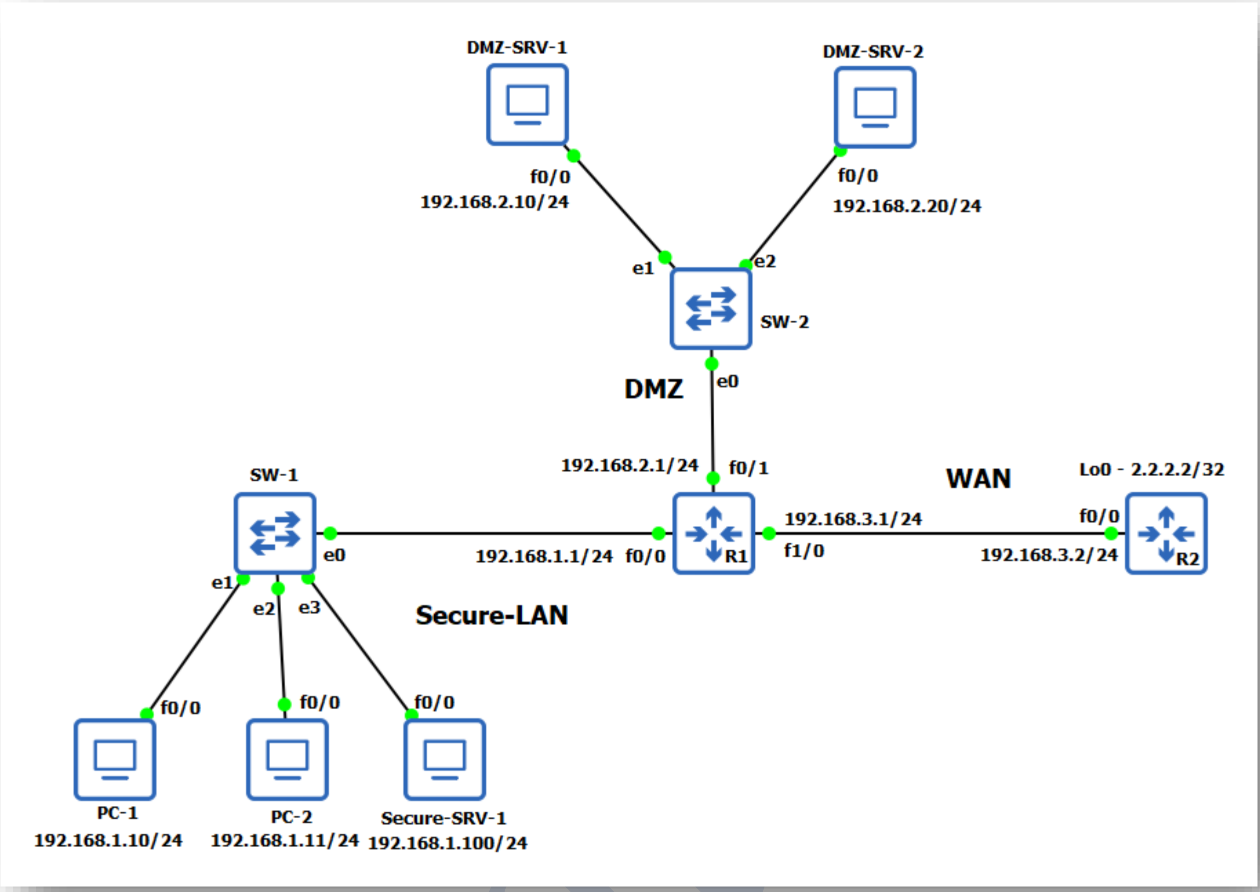
# Time-Range ACL




## Task

1. Configure routers R1 & R2 with IP address as shown in topology and configure enable password as **ccna**. **(Refer Lab-16 for task 1 to 3)**
2. Configure routers as Host (PC / Servers)
3. Configure default routing to provide connectivity between them.
4. Configure HTTP service on R2.
5. Configure R2 as NTP server and R1 as NTP client
6. Configure time range ACL to block internet - http service (R2's loopback IP ) during the office working hours (9 am to 6 pm)

## Task-4: Configure HTTP service on R2.

R2 Configuration:

R2#config t
R2(config)#enable password ccna
R2(config)#username user1 secret pass1
R2(config)#ip http server
R2(config)#ip http secure-client-auth
R2(config)#line vty 0 15
R2(config-line)#login local

R2(config-line)#transport input ssh telnet
R2(config-line)#exit
R2(config)#exit
R2#

- ✓ **Verification & Testing:**

**PC-1#ping 2.2.2.2**
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/332/1164 ms

**PC-1#telnet 2.2.2.2 80**
Trying 2.2.2.2, 80 ... Open

PC-1#

**PC-2#ping 2.2.2.2**
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 100/333/1076 ms

**PC-2#telnet 2.2.2.2 80**
Trying 2.2.2.2, 80 ... Open

PC-2#

**Secure-SRV-1#ping 2.2.2.2**
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/327/1108 ms

**Secure-SRV-1#telnet 2.2.2.2 80**
Trying 2.2.2.2, 80 ... Open

Secure-SRV-1#

## Task-5: Configure R2 as NTP server and R1 as NTP client

**R2 Configuration:**

R2#config t
R2(config)#clock timezone IST +5 30
R2(config)#ntp master
R2(config)#exit
R2#clock set 17:45:00 6 Nov 2020

**R1 Configuration:**

R1#config t
R1(config)#clock timezone IST +5 30
R1(config)#ntp server 192.168.3.2
R1(config)#exit
R1#

- ✓ **Verification & Testing:**

**R2#sh ntp status**
Clock is synchronized, stratum 8, reference is 127.127.1.1
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**18
ntp uptime is 65400 (1/100 of seconds), resolution is 4000
reference time is E34FBD04.591AD8F9 (17:46:04.348 IST Fri Nov 6 2020)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 437.92 msec, peer dispersion is 437.69 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
system poll interval is 16, last update was 16 sec ago.
R2#

**R1#sh ntp status**
Clock is synchronized, stratum 9, reference is 192.168.3.2
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**18
ntp uptime is 64400 (1/100 of seconds), resolution is 4000
reference time is E34FBD43.2C9CF91C (17:47:07.174 IST Fri Nov 6 2020)
clock offset is -1.8256 msec, root delay is 28.40 msec
root dispersion is 7942.08 msec, peer dispersion is 937.54 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000000000 s/s
system poll interval is 64, last update was 7 sec ago.
R1#

**R2#sh clock**
17:49:12.199 IST Fri Nov 6 2020
R2#

**R1#sh clock**
17:49:28.021 IST Fri Nov 6 2020
R1#

**Task-6: Configure time range ACL to block all access to R2's loopback IP (Internet) during the office working hours (9 am to 6 pm)**

**R1 Configuration:**

R1#config t
R1(config)#time-range Office-Hours
R1(config-time-range)#periodic weekdays 09:00 to 18:00
R1(config-time-range)#exit
R1(config)#time-range Internet-Access
R1(config-time-range)#periodic weekdays 00:00 to 23:59

R1(config-time-range)#exit
R1(config)# access-list 100 permit ip host 192.168.1.100 any time-range Office-Hours
R1(config)#access-list 100 deny ip any any time-range Office-Hours
R1(config)#access-list 100 permit ip any any time-range Internet-Access
R1(config)#int fa1/0
R1(config-if)#ip access-group 100 out
R1(config-if)#exit
R1(config)#exit
R1#

- ✓ **Verification & Testing:**

**R1#sh clock**
17:50:15.307 IST Fri Nov 6 2020
R1#

**R1#sh access-lists**
Extended IP access list 100
   10 permit ip host 192.168.1.100 any time-range Office-Hours (active) (2 matches)
   20 deny ip any any time-range Office-Hours (active) (15 matches)
   30 permit ip any any time-range Internet-Access (active)
R1#

**R1#sh time-range**
time-range entry: Internet-Access (active)
  periodic weekdays 0:00 to 23:59
  used in: IP ACL entry
time-range entry: Office-Hours (active)
  periodic weekdays 9:00 to 18:00
  used in: IP ACL entry
  used in: IP ACL entry
R1#

**PC-1#ping 2.2.2.2**
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)

**PC-1#telnet 2.2.2.2 80**
Trying 2.2.2.2, 80 ...
% Destination unreachable; gateway or host down

PC-1#

**PC-2#ping 2.2.2.2**
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)

**PC-2#telnet 2.2.2.2 80**
Trying 2.2.2.2, 80 ...
% Destination unreachable; gateway or host down

PC-2#

**Secure-SRV-1#ping 2.2.2.2**
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/40/44 ms

**Secure-SRV-1#telnet 2.2.2.2 80**
Trying 2.2.2.2, 80 ... Open

Secure-SRV-1#

**R1#sh access-lists**
Extended IP access list 100
  10 permit ip host 192.168.1.100 any time-range Office-Hours (active) (10 matches)
  20 deny ip any any time-range Office-Hours (active) (33 matches)
  30 permit ip any any time-range Internet-Access (active)
R1#

**Now wait for office hours to complete.**

**R1#sh clock**
17:57:28.792 IST Fri Nov 6 2020
R1#

**R1#sh clock**
18:01:09.930 IST Fri Nov 6 2020
R1#

**R1#sh access-lists**
Extended IP access list 100
  10 permit ip host 192.168.1.100 any time-range Office-Hours (inactive) (14 matches)
  20 deny ip any any time-range Office-Hours (inactive) (34 matches)
  30 permit ip any any time-range Internet-Access (active)
R1#

**PC-1#ping 2.2.2.2**
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/42/44 ms

**PC-1#telnet 2.2.2.2 80**
Trying 2.2.2.2, 80 ... Open

PC-1#



info@rsatechforum.in

**PC-2#ping 2.2.2.2**
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/40/44 ms

**PC-2#telnet 2.2.2.2 80**
Trying 2.2.2.2, 80 ... Open

PC-2#

**Secure-SRV-1#ping 2.2.2.2**
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 36/40/44 ms

**Secure-SRV-1#telnet 2.2.2.2 80**
Trying 2.2.2.2, 80 ... Open

Secure-SRV-1#

**R1#sh access-lists**
Extended IP access list 100
  10 permit ip host 192.168.1.100 any time-range Office-Hours (inactive) (14 matches)
  20 deny ip any any time-range Office-Hours (inactive) (34 matches)
  30 permit ip any any time-range Internet-Access (active) (35 matches)
R1#

## To Remove ACL

R1#config t
R1(config)#no access-list 100
R1(config)#int fa1/0
R1(config-if)#ip access-group 100 out
R1(config-if)#exit
R1(config)#no time-range Office-Hours
R1(config)#no time-range Internet-Access
R1(config)#exit
R1#

**Important Commands:**

sh access-lists
sh ip access-lists
sh run | sec access-list
sh time-range



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